# Exit Ticket · reflect and retrieve

## Exit Ticket

Knowing a topic name alone proves that you can apply the associated algorithm.

## Use key vocabulary from memory

1. Explain synoptic in your own words.

2. Explain retrieval in your own words.

3. Explain misconception in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 304 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Final specification and synoptic checkpoint

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Name a PageRank property beyond counting inbound links.

### Q2

What is a page fault?

### Q3

What does an INNER JOIN return?

### Q4

Search value 11 in [1,3,5,7,9,11,13], showing mid indexes.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

### Core idea 1

Use evidence rather than ticks alone — A topic is secure when a learner can solve an unfamiliar example, not merely recognise its name.

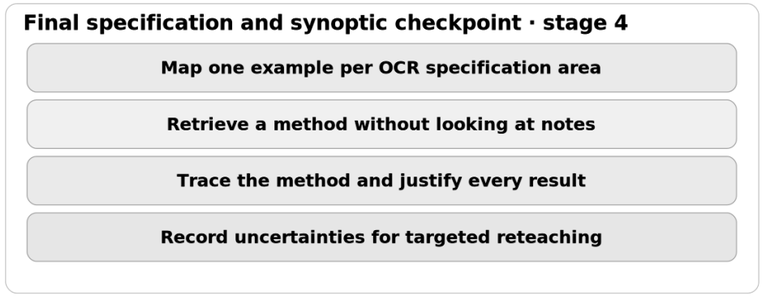
### Core idea 2

Find links between papers — Paper 1 numerical representation and database knowledge can underpin Paper 2 programming and algorithm questions.

### Core idea 3

Prioritise gaps — Record the exact misconception or missing step so later Paper 1/2 reteaching sessions target observable weaknesses.

**Key visual · complete concept**




### Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

### Question 3

A school chooses AI scoring for pupil support. Give three contextual safeguards and explain risks. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

State roles of PC, MAR, MDR, CIR and ACC, plus one bus role. [6]

## Question 2

A sorted list [2,4,6,8,10,12,14] is searched for 12. Trace binary search, then explain worst-case Big O. [6]





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

synoptic — Combining knowledge across topics.

retrieval — Reproduce knowledge from memory.

misconception — Incorrect conceptual understanding.

intervention — Specific teaching targeting a gap.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Evaluate the claim 'all sorts take O(n log n) in the worst case'.

## Follow the worked method

1. Predict first — Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.

2. Check reasoning — Merge sort has O(n log n) worst-case growth and generally needs O(n) auxiliary memory.

**Source model · final illustrated step**

**IDO · step 2**

**Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.**

**Merge sort has O(n log n) worst-case growth and generally needs O(n) auxiliary memory.**

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first
Identify exactly which implementation state is wrong, e.g. high never decreases.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.